

Structure versus Randomness: A Computational Synthesis of Additive Number Theory, Machine Learning, and Cryptography

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Abstract

Two companion papers examined, respectively, how recurrence in dynamical systems yields theorems of additive number theory, and how matrix theory yields the algorithms of big-data machine learning. This paper argues that both, together with modern cryptography, are facets of a single organizing principle--the *structure-versus-randomness dichotomy*--and that three concrete mathematical instruments recur across all three fields: character (exponential) sums, random linear maps, and spectral gaps. We make each bridge computational in an open-source Python package, *crypto-synthesis*. Character sums, which certify equidistribution of $n\alpha$ and $n^2\alpha$ in the number-theory project, here certify the pseudorandomness of Legendre sequences via the Weil bound. Random linear maps, which preserved geometry through Johnson-Lindenstrauss projections in the machine-learning project, here purify weak randomness through the leftover hash lemma. Spectral gaps, which drove spectral clustering and PageRank, here drive expander mixing and Cayley-graph hashing. Finally, Learning With Errors--the foundation of post-quantum cryptography--is exhibited as the machine-learning project's linear algebra with a single noise term added, and we demonstrate that removing that term collapses the problem to Gaussian elimination. All parameters are deliberately pedagogical; the contribution is conceptual and expository, not an attack on any deployed system.

1 Introduction

Terence Tao has described much of modern mathematics as governed by a dichotomy between *structure* and *randomness* [10]: many objects are either algebraically structured, and hence analysable by the tools of that structure, or pseudorandom, and hence analysable by the tools of probability--with the deepest theorems living at the interface, showing that an arbitrary object decomposes into a structured part and a pseudorandom error. This paper takes the dichotomy as a literal bridge between three fields that are usually taught apart.

The two companion projects established one half of the picture each. The first showed that recurrence--orbits of measure-preserving systems returning near their starting

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points--is equivalent, through the Furstenberg correspondence principle, to the existence of arithmetic structure in dense sets of integers (Szemerédi's theorem and its polynomial refinements). The second showed that the compressibility of data, the preservation of geometry under random projection, the connectivity of graphs, and the convergence of iterative eigensolvers are all read off the singular values and eigenvalues of a matrix, and that these facts *are* the algorithms of large-scale machine learning. The present paper closes the triangle: cryptography is the engineering of the pseudorandom side of the dichotomy, and it reuses, almost verbatim, the same three instruments.

instrument	appears in ML / number theory as	appears in crypto as
character sums	Weyl equidistribution	pseudorandom sequences
random linear maps	Johnson-Lindenstrauss	randomness extractors
spectral gap	Fiedler value, PageRank	expander hashing
noisy linear algebra	least squares, SVD	Learning With Errors

Sections 2-6 develop the four bridges; Section 7 reports the experiments in the accompanying package; Section 8 reflects on the unity and states the honest limitations of a computational treatment.

2 Learning With Errors: noisy linear algebra

Definition 2.1 (LWE [7]). Fix a modulus q , a dimension n , and an error distribution χ on $\mathbb{Z}_q$ concentrated on small values. Given $A \in \mathbb{Z}_q^{m \times n}$ chosen uniformly and $b = As + e \pmod{q}$ with secret $s \in \mathbb{Z}_q^n$ and $e \leftarrow \chi^m$, the *search-LWE* problem is to recover s .

The pivotal observation is what happens when $\chi = 0$.

Proposition 2.2. *If $e = 0$ then s is recoverable in polynomial time by Gaussian elimination over $\mathbb{Z}_q$ whenever A has rank n .*

Proposition 2.2 is the linear algebra of the machine-learning project used as a cryptanalytic attack; it succeeds instantly. The security of LWE is precisely the assertion that the small perturbation e destroys this solvability. Regev's theorem makes the assertion rigorous by reduction from worst-case lattice problems.

Theorem 2.3 (Regev [7], informal). *For appropriate parameters, an efficient algorithm solving average-case LWE yields an efficient (quantum) algorithm approximating the worst-case shortest vector problem GapSVP and the shortest independent vectors problem SIVP to within polynomial factors.*

Because GapSVP is believed hard even for quantum computers, LWE underpins post-quantum schemes such as Kyber. On the constructive side, the same equations give encryption: a public key (A, b) with $b = As + e$ encrypts a bit μ as $(a, c) = (\sum_{i \in R} A_i, \sum_{i \in R} b_i + \mu \lfloor q/2 \rfloor)$ for a random row subset R , and decryption tests whether $c - \langle a, s \rangle$ is closer to 0 or to $q/2$. The package implements this and confirms correct round-trip decryption (Section 7).

3 Lattices: the geometry behind the algebra

The dual view of LWE is geometric. A lattice $\Lambda = \{Bx : x \in \mathbb{Z}^n\}$ has a basis B whose quality is measured by the *orthogonality defect* $\delta(B) = \prod_i \|b_i\| / |\det B|$, equal to 1 exactly when B is orthogonal. The Gram-Schmidt process that computes the defect is the QR decomposition of the machine-learning project in number-theoretic notation. The LLL algorithm [6] reduces the defect in polynomial time.

Theorem 3.1 (LLL [6]). *Given a basis of an n -dimensional lattice, the LLL algorithm outputs, in polynomial time, a basis $b_1, \dots, b_n$ with $\|b_1\| \leq 2^{(n-1)/2} \lambda_1(\Lambda)$, where λ_1 is the length of the shortest nonzero lattice vector.*

Embedding an LWE instance into a lattice so that the short vector $(e, 1)$ is a lattice point (the Kannan embedding) turns secret recovery into short-vector search; LLL surfaces it at small parameters. This is the exact complement of Proposition 2.2: what algebra could not extract from the noisy system, geometry can, and the arms race between embeddings and reduction algorithms is the quantitative heart of lattice cryptanalysis.

4 Character sums: certifying pseudorandomness

In the number-theory project, Weyl's criterion converted equidistribution of $n\alpha$ and $n^2\alpha$ into decay of exponential sums. In cryptography the same principle certifies that number-theoretic sequences look random. Write $\left(\frac{\cdot}{p}\right)$ for the Legendre symbol modulo a prime p .

Theorem 4.1 (Weil bound [1]). *Let χ be a nontrivial multiplicative character modulo p and $f \in \mathbb{F}_p[x]$ of degree d that is not, up to scalar, a perfect $\text{ord}(\chi)$ -th power. Then*

$$\left| \sum_{x \in \mathbb{F}_p} \chi(f(x)) \right| \leq (d-1)\sqrt{p}.$$

The square-root cancellation is the pseudorandomness: the character values $\chi(f(x))$ behave like independent random signs. For $f(x) = x^2 + a$ with $a \neq 0$ the bound reads $|\sum| \leq \sqrt{p}$; the excluded case $a = 0$, where x^2 is a perfect square and the sum degenerates to $p-1$, is a vivid reminder that the theorem is a statement about the *absence of structure*. Consequences include the balance of quadratic residues and the low autocorrelation of the Legendre sequence $\left(\left(\frac{n}{p}\right)\right)_n$, the basis of Legendre pseudorandom generators [5]. The multiplicative analogue--equidistribution of the discrete-logarithm orbit $\{g^k \bmod p\}$ --quantifies the scrambling of exponentiation and is measured, as in the number-theory project, by star discrepancy.

5 Random linear maps: the leftover hash lemma

The machine-learning project used random linear maps to preserve geometry (Johnson-Lindenstrauss). Cryptography uses them to preserve (indeed manufacture) uniformity. The min-entropy of a source X is $H_\infty(X) = -\log_2 \max_x \Pr[X = x]$.

Theorem 5.1 (Leftover hash lemma [4]). *Let $\mathcal{H} = \{h : \{0,1\}^n \rightarrow \{0,1\}^L\}$ be a universal family and X a source with $H_\infty(X) \geq k$. For h drawn uniformly from $\mathcal{H}$, the pair $(h, h(X))$ is ε -close in statistical distance to (h, U_L) with*

$$\varepsilon \leq \frac{1}{2} \sqrt{2^{L-k}}.$$

A random binary matrix $M \in \mathbb{F}_2^{L \times n}$ acting by $x \mapsto Mx$ is a linear universal hash--a Johnson-Lindenstrauss sketch over the field $\mathbb{F}_2$ --and hence a strong extractor. Theorem 5.1 says that to harvest L near-uniform bits one needs an entropy surplus of $2 \log(1/\varepsilon)$ over L . The kinship with the geometric JL lemma is not merely cosmetic: both are concentration statements for random linear maps, one in ℓ_2 geometry and one in total-variation distance. The package verifies the bound empirically and also implements the inner-product two-source extractor $\langle x, y \rangle \bmod 2$, whose bias analysis is a slice of additive combinatorics over $\mathbb{F}_2^n$, linking back to the number-theory project.

6 Spectral gaps: expander mixing and hashing

The machine-learning project extracted cluster structure and PageRank runtimes from the second eigenvalue of a graph operator. Cryptography extracts pseudorandomness from the same quantity. For a d -regular graph G on N vertices, let λ be the second-largest eigenvalue of the adjacency matrix in absolute value; the *spectral gap* is $d - \lambda$.

Theorem 6.1 (Expander mixing lemma [3]). *For all vertex subsets S, T ,*

$$\left| e(S, T) - \frac{d}{N} |S| |T| \right| \leq \lambda \sqrt{|S| |T|}.$$

A small λ (large gap) forces the edge distribution to imitate a random graph, and the random-walk mixing time is $O(\frac{1}{d-\lambda} \log N)$ --the very rate at which power iteration converged in the machine-learning project. Expander- and Cayley-graph hash functions [8] exploit exactly this: a message drives a walk on an expander, and expansion guarantees that distinct messages land on well-separated vertices, yielding collision resistance. Optimal expanders (Ramanujan graphs, $\lambda \leq 2\sqrt{d-1}$) are furnished by the Lubotzky-Phillips-Sarnak construction [2], itself built from deep number theory--so the triangle closes on itself.

7 The computational companion

7.1 Package

crypto-synthesis (Python, NumPy only, MIT licence) has five modules--lwe, lattice, characters, extractors, expander--each implementing one bridge, with seventeen unit tests binding code to the theorems above. The LWE attack, for instance, is Gaussian elimination over $\mathbb{Z}_q$:

```
inv = pow(int(M[row, col]), -1, q) # modular pivot inverse
M[row] = (M[row] * inv) % q
for r in range(m): # eliminate column 'col'
    if r != row and M[r, col]:
        M[r] = (M[r] - M[r, col] * M[row]) % q
```

which recovers s from noiseless samples and provably cannot from noisy ones.

7.2 Experiments

LWE (Section 2). With $n = 15$, $m = 40$, $q = 257$, Gaussian noise $\sigma = 2$: the noiseless system $b = As$ yields s exactly by elimination; the identical attack on $b = As + e$ (error entries up to 8) fails; and Regev encryption of an eight-bit message decrypts correctly.

Lattices (Section 3). LLL reduces a badly skewed two-dimensional basis from orthogonality defect 2545.5 to 1.118. On a Kannan-embedded LWE lattice with $n = 4$, $q = 97$, the shortest reduced vectors have norms $\{8.0, 8.4, 8.6\}$, comparable to the true error norm $\|e\| = 3.16$ --the error surfaces as a short vector.

Character sums (Section 4). For $p = 1009$, the sums $\sum_n \left(\frac{n^2+a}{p}\right)$ equal -1 for $a \in \{1, 7, 42\}$ against the ceiling $\sqrt{p} \approx 31.8$ (square-root cancellation), while the degenerate case $a = 0$ gives 1008, exactly as Theorem 4.1 predicts for a perfect square. The Legendre sequence has maximum autocorrelation 0.004 over shifts 1-19, and the discrete-log orbit $g^k \bmod p$ has star discrepancy 0.0009.

Extractors (Section 5). A source of 40 bits each biased to 0.35 has min-entropy $k \approx 24.9$; a random $\mathbb{F}_2$ -linear map to $L = 8$ bits yields empirical statistical distance 0.011 from uniform, comfortably inside the regime where the leftover-hash-lemma bound $\frac{1}{2}\sqrt{2^{L-k}} \approx 0.0015$ is small.

Expanders (Section 6). For the Cayley graph of $\mathbb{Z}/101$ with generators $\{\pm 1, \pm 10, \pm 30\}$ ($d = 6$), the second eigenvalue is $\lambda \approx 5.59$; the expander mixing lemma holds for every random subset pair tested ($|e(S, T) - d|S||T|/N| = 42.3 \leq \lambda\sqrt{|S||T|} = 193.5$ in a representative case), and single-bit message changes move the Cayley hash to unrelated vertices. As Section 8 notes, this small circulant is a mediocre expander--a deliberate, honest choice at pedagogical scale.

8 Discussion

The unity. The three projects are one subject. A set of integers, a data matrix, and a cipher's output stream are all interrogated with the same question--structured or random?--and answered with the same three tools. Weyl's criterion, the Johnson-Lindenstrauss lemma, and the spectral gap are not analogies across the fields; they are the same theorems, deployed for different ends. Learning With Errors makes the point unmissable: it is literally the linear algebra of data science with a noise term whose sole purpose is to move the problem from the structured side of the dichotomy to the pseudorandom side.

Honest limitations. A computational companion verifies and illustrates; it does not prove, and at cryptographic scale it must not pretend to. Every parameter here is a toy: LWE dimensions in the teens, primes near 10^3 , graphs on a hundred vertices. These choices make the mathematics visible and give no leverage against real systems, whose security parameters are larger by orders of magnitude and whose algebraic structure is chosen to defeat exactly the textbook methods shown here. Two caveats deserve emphasis:

the circulant Cayley graphs used are far from Ramanujan, so their mixing is slow; and the lattice demonstrations recover short vectors in dimensions where essentially any method succeeds. The value is in the bridge, not the break.

Extensions. The natural next steps continue the trilogy’s method: Gowers uniformity norms as a common quantitative measure of pseudorandomness across all three fields; the number-theoretic construction of Ramanujan graphs, closing the loop from cryptography back to the modular forms adjacent to the number-theory project; and a treatment of the dual attack surface, where machine-learning regression is itself used as a cryptanalytic tool against structured noise.

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